

Batocera v42 Audio Module

Setup & Troubleshooting Reference

Raspberry Pi 5 + PCM5102 DAC + PAM8403 Amp + PJ-359B-5P Jack

GPIO wiring - 5 wires only

DAC Pin	Pi 5 Physical Pin	BCM Number	Signal
VIN	Pin 2	—	5V power
GND	Pin 6	—	Ground
BCK	Pin 12	BCM 18	I2S bit clock
LCK	Pin 35	BCM 19	I2S left/right select
DIN	Pin 40	BCM 21	I2S data
SCK	—	—	No wire - handled by front pad bridge

Jumper configuration (back and front of DAC board)

Before wiring, confirm the solder bridges on the back of the board are set correctly. Do not add external wires to any pin that already has a solder bridge — this causes conflicts.

Jumper	Function	Correct state	Pads bridged
H1L	FLT - filter select	Normal latency	Middle + right

H2L	DEMP - de-emphasis	Off	Middle + right
H3L	XSMT - soft mute	Unmuted (HIGH)	Middle + LEFT
H4L	FMT - audio format	I2S	Middle + right
SCK	SCK - internal bit clock		Two pads in front of SCK

SCK — front pad bridge

The front of the PCM5102 board has a dedicated pair of pads near the SCK pin. Bridge these two pads with solder to tie SCK to GND internally. This tells the chip to generate its own system clock via the internal PLL from BCK. The Pi 5 does not supply a master clock.

NOTE: If the front SCK pads are bridged, do NOT also wire the SCK header pin to GND. Pick one method only.

1. Batocera Configuration

1.1 /boot/config.txt

Edit this file by mounting the SD card on PC or via SSH. Add or confirm these lines:

```
nano /boot/config.txt
dtparam=audio=off
dtoverlay=hifiberry-dac
dtoverlay=vc4-kms-v3d,noaudio
```

WARNING: Do not have both `dtoverlay=vc4-kms-v3d` and `dtoverlay=vc4-kms-v3d,noaudio` in the file. Keep only the `noaudio` version. Two entries cause a conflict.

NOTE: Do not add `dtparam=i2s=on` — the `hifiberry-dac` overlay handles this automatically.

NOTE: Do not create `/etc/asound.conf` — Batocera v42 uses Pipewire, not ALSA directly. `asound.conf` conflicts with Pipewire.

1.2 /userdata/system/batocera.conf

Add or confirm these lines in the audio section:

```
nano /userdata/system/batocera.conf
```

```
audio.device=hw:0,0  
audio.volume=70
```

NOTE: In Batocera v42 on Pi 5, HiFiBerry may not appear in the Audio Output UI. This is a known Pipewire routing quirk. Setting hw:0,0 directly works correctly even if the UI shows Built-in Audio Stereo as the sink.

2. Verification & Testing

2.1 After first boot — check DAC is detected

```
| aplay -l
```

Expected output includes:

```
| card 0: sndrpihifiberry [snd_rpi_hifiberry_dac], device 0: HiFiBerry  
| DAC HiFi pcm5102a-hifi-0
```

If card not shown, check:

- config.txt edits saved correctly
- No duplicate dtoverlay=vc4-kms-v3d lines
- DAC physically connected and powered
- Run: `dmesg | grep -i hifi`

2.2 Check Pipewire routing

```
| wpctl status
```

Under Streams, the game audio should show:

```
| output_FL > HifiBerry DAC HiFi pcm5102a-hifi-0:playback_FL [active]  
| output_FR > HifiBerry DAC HiFi pcm5102a-hifi-0:playback_FR [active]
```

Monitor audio performance in real time while game is running:

```
| pw-top
```

Check ERR column — should stay at 0. Increasing ERR values indicate buffer underruns.

2.3 Hardware test — bypass Pipewire

To test DAC hardware directly without Pipewire interference:

```
| /etc/init.d/S06audio stop  
| aplay -D hw:0,0 -f S32_LE -r 48000 -c 2 /dev/urandom
```

Should hear clean white noise/hiss from earphones. Press Ctrl+C after 2-3 seconds.

- Clean white noise = DAC hardware working correctly
- Rhythmic buzzing = electrical interference — check wiring and jumper bridges
- Silence = DAC not receiving signal — check I2S pin connections
- Device busy error = Pipewire still running, stop it first

2.4 Check active stream sample rate

```
| cat /proc/asound/card0/pcm0p/sub0/hw_params
```

Expected output while audio is playing:

```
| format: S32_LE  
| rate: 48000  
| channels: 2
```

2.5 Check for Pi throttling

```
| vcgencmd measure_temp  
| vcgencmd get_throttled
```

Temperature should stay below 80C during gameplay. throttled=0x0 means no throttling events. Any non-zero value indicates under-voltage or thermal issues.

3. Common Problems & Fixes

Symptom	Cause	Fix
No audio at all, DAC detected	H3L jumper on wrong side (XSMT low)	Re-bridge H3L to middle+left pads
Distorted bass-heavy audio	XSMT wire conflicting with solder bridge	Remove external XSMT wire, use solder bridge only
Rhythmic buzzing at idle	Electrical interference on I2S lines	Check for stray wire strands, verify SCK bridge on front pad
Audio works but hums at idle	Ground loop	Verify star ground — all GNDs meet at one point
Pi reboots when amp connected	Amp GND sharing Pi ground path	Route amp GND directly to power bank B- only
Amp cuts off after 15-20 seconds	Power bank USB-A auto-shutoff	Bypass USB-A, solder to C15/C16 and B- pads directly
One ear only with earphones	TRRS plug in TRS jack mismatch	Use TRRS-to-TRS adapter or replace jack with PJ-320A
HiFiBerry not in Batocera UI	Pipewire quirk on Pi 5 Batocera v42	Set audio.device=hw:0,0 in batocera.conf, audio still works
aplay: device or resource busy	Pipewire holding device	Run /etc/init.d/S06audio stop first

No soundcards found in aplay	Conflicting dtoverlay lines	Remove duplicate vc4-kms-v3d line in config.txt
Occasional volume dip in games	Pipewire sample rate conversion	Audio routes correctly to HiFiBerry, ERR=0 means no real issue

4. Quick Reference — Full Component Checklist

Before powering on after a fresh flash, verify all of these:

Item	Status to confirm
H1L	Middle + right bridged
H2L	Middle + right bridged
H3L	Middle + LEFT bridged (XSMT high)
H4L	Middle + right bridged
SCK front pad	Two pads bridged with solder
No XSMT external wire	Only solder bridge used, no jumper wire
VIN → Pi Pin 2	5V, not 3.3V
GND → Pi Pin 6	Confirmed
BCK → Pi Pin 12	BCM 18
LCK → Pi Pin 35	BCM 19
DIN → Pi Pin 40	BCM 21
100 ohm resistors	One on L channel, one on R channel
10uF caps	– leg toward DAC, + leg toward amp/jack
1000uF cap on PAM8403	+ to 5V pad, – to GND pad
Amp power from C15/C16	Not from USB-A port
Amp GND to B–	Not connected to Pi Pin 6 or DAC GND
Speaker – not to GND	Each speaker only between + and – pads
config.txt — hifiberry-dac	dtoverlay=hifiberry-dac present
config.txt — audio off	dtparam=audio=off present
config.txt — noaudio	dtoverlay=vc4-kms-v3d,noaudio (no duplicate)
batocera.conf	audio.device=hw:0,0